Hadi Arbabi

+44 (0) 114 222 5751
h.arbabi@sheffield.ac.uk
researchgate.net/profile/Hadi_Arbabi

Hadi's work and research interests sit at the interface of data-driven urban analytics and planning. Their doctoral thesis focused on a spatially multi-scale examination of urban systems and the extent to which their performance is influenced by their embedded physical infrastructure. Hadi's broader research activity and interests include but are not limited to practical uses of urban scaling and allometry, city morphology, infrastructure planning for agglomeration, and network analysis of intra- and inter-city urban flows. As of December 2021, Hadi has over 60 citations and an *h*-index of 5.

Education

The University of Sheffield

PhD 2015 – 2019
Resources, Infrastructure Systems, & built-Environments,

- Thesis Title:
  Urban Productivity & Spatial Patterns Across Scales–
  A Multi-Scale Exploration of Urban Networks and Their Hierarchical Configurations

MEng 2011 – 2015
Architectural Engineering Design (First-Class Honors),

- Dissertation Title:
  Influence of Anthropic and Spatial Characteristics of Cities upon Their Energy Metabolism

Academic Positions

The University of Sheffield

Lecturer August 2020 – Present
Resources, Infrastructure Systems, & built-Environments Research Group

Post-Doctoral Research Associate April 2019 – July 2020
Digital Urban Characterization [EP/S016627/1, EP/V012053/1]

Part-Time Research Associate November 2018 – February 2019
Engineering Complexity Resilience Network Plus [EP/N010019/1]

Professional Memberships

Regional Studies Association (no. 15429)
Regional Sciences Association International (no. RSAI08605)

Review Activities

Regional Studies, Journal of the Royal Society Interface, Science of the Total Environment, Journal of Environmental Management, ISPRS International Journal of Geo-Information, Energies, Sustainability, Applied Sciences

Professional Standing & Awards

'Real Zero' in a Hurry, DecarboN8 International Conference Summer 2021
Invited Session Chair for Innovation and Infrastructure Sustainability: Methods, Metrics and Measures

International Industrial Ecology Day — Summer 2021
Session Organizer for Bottom-Up Urban Built Environment MFA: Data, Methods, Challenges and Comparability

The University of Sheffield — Autumn 2015 – Summer 2019
Faculty scholarship to undertake doctoral research

Regional Science Association International, British and Irish Section — Summer 2017
The Early Career Prize for Best Presentation — (Paper C6)

The University of Sheffield — Summer 2015
Mappin Medal and Premium for Greatest Distinction Shown by a Candidate on a MEng Program

The Institution of Structural Engineers (IStructE) — Spring 2014 – Winter 2015
Young Members' Panel invited membership

Chartered Institute of Building Services Engineers (CIBSE), Yorkshire Region — Autumn 2013
CIBSE Yorkshire Award for Outstanding Academic Achievement

Research Articles

J9. A Scalable Data Collection, Characterization, and Accounting Framework for Urban Material Stocks. *Journal of Industrial Ecology*, (2021), with M. Lanau, X. Li, G. Meyers, M. Dai, M. Mayfield, & D. Densley Tingley. doi:10.1111/jiec.13198

J8. Mapping Resource Effectiveness across Urban Systems. *npj Urban Sustainability*, **1**, 20 (2021), with L. M. Tan, P. Brockway, D. Densley Tingley, & M. Mayfield. doi:10.1038/s42949-020-00009-3

J7. On the Use of Random Graphs in Analysing Resource Utilization in Urban Systems. *Royal Society Open Science*, **7**, 200087 (2020), with G. Punzo, G. Meyers, L. M. Tan, Q. Li, D. Densley Tingley, & M. Mayfield. doi:10.1098/rsos.200087

J6. Productivity, Infrastructure, and Urban Density – an Allometric Comparison of Three European City-Regions across Scales. *Journal of the Royal Statistical Society: Series A (Statistics in Society)*, **183**(1), 211-228 (2020), with M. Mayfield & P. McCann. doi:10.1111/rssa.12490

J5. Urban Performance at Different Boundaries in England and Wales through the Settlement Scaling Theory. *Regional Studies*, **53**(6), 887-899 (2019), with M. Mayfield & G. Dabinett. doi:10.1080/00343404.2018.1490501

J4. On Development Logic of City-Regions: Inter- Versus Intra-City Mobility in England and Wales. *Spatial Economic Analysis*, **14**(3), 301-320 (2019), with M. Mayfield & P. McCann. doi:10.1080/17421772.2019.1569762

J3. An Ecological Thermodynamic Approach in Urban Metabolism: Measuring Resource Utilization with Open System Network Effectiveness Analysis. *Applied Energy*, **254**, 113618 (2019), with L. M. Tan, P. E. Brockway, D. Densley Tingley, & M. Mayfield. doi:10.1016/j.apenergy.2019.113618

J2. Ecological Network Analysis on Intra-city Metabolism of Functional Urban Areas in England and Wales. *Resources, Conservation & Recycling*, **138**, 172-182 (2018), with L. M. Tan, Q. Li, Y. Sheng, D. Densley Tingley, M. Mayfield, & D. Coca. doi:10.1016/j.resconrec.2018.06.010

J1. Urban and Rural — Population and Energy Consumption Dynamics in Local Authorities within England and Wales. *Buildings*, **6**(3), 34 (2016), with M. Mayfield. doi:10.3390/buildings6030034

Chapters & Reports

R2. Resource Effectiveness in and across Urban Systems. In R. C. Brears (Ed.) *The Palgrave Encyclopedia of Urban and Regional Futures*, Cham: Palgrave Macmillan, (2021), with L. M. Tan. doi:10.1007/978-3-030-51812-7_202-1

R1. Quantifying Agglomeration Productivity Potential in Long-Term Infrastructure Planning (PIN – Productivity Projects Fund). *Productivity Insights Network*, (2020), with J. Pannell, S. Hincks, & G. Punzo. doi: available here

Others

O2. Comments on Bettignies et al., 'The Scale-Dependent Behaviour of Cities: A Cross-Cities Multi-scale Driver Analysis of Urban Energy Use' Sustainability 2019, 11, 3246. *Sustainability*, (Under Review), with G. Meyers, L. M. Tan, & M. Mayfield.

O1. Comments on 'A Multi-Level Framework for Metabolism in Urban Energy Systems from an Ecological Perspective' by Barrera et al. (2018). *Resources, Conservation & Recycling*, 136, 463-465 (2018), with L. M. Tan & M. Mayfield. doi:10.1016/j.resconrec.2018.04.028

Invited Talks

T2. The Tensions between Density, Mobility, & Economic Output in Long-Term Mobility Planning. In *Northern Evidence Academic Forums.* Transport for the North, UK: 12 May 2021.

T1. On the Development Logic of City-Regions. In *Connect to Collaborate: Accelerating Infrastructure Transitions towards Connected Places.* The University of Bristol, UK: 23 May 2019. (An ESRC co-funded workshop through the Productivity Insights Network)

Conferences

C10. Spatial Consistency and Patterns in Agglomeration Elasticities within and across European Urban Systems. In *Regional Science Association International-British and Irish Section 47th Annual Conference.* Cambridge, UK: 2019.

C9. Limiting Factors in Achieving Overall Circular Economic Effectiveness: an Analytical Assessment. In *1st International Conference on Construction Circular Economy.* Manchester, UK: 2019, with G. Punzo, G. Meyers, L. M. Tan, Q. Li, D. Densley Tingley, & M. Mayfield.

C8. Finding Northern Powerhouse, or How to Grow Your Own City Regions. In *Regional Studies Association Winter Conference.* London, UK: 2017, with M. Mayfield & P. McCann.

C7. Ecological Network Analysis on Intra-city Metabolism of Functional Urban Areas in England and Wales. In *Spaces & Flows: Eighth International Conference on Urban and ExtraUrban Studies.* Hull, UK: 2017, with L. M. Tan, Q. Li, Y. Sheng, N. Stanton, T. J. Valentine, & E. B. Fraser.

C6. A Look at Urban Performance Balance across Different Boundaries in England and Wales. In *Regional Science Association International-British and Irish Section 46th Annual Conference.* Harrogate, UK: 2017, with M. Mayfield, & G. Dabinett.

C5. The Importance of Understanding the Material Metabolism of the Built Environment. In *World Sustainable Built Environment Conference.* Hong Kong: 2017, with D. Densley Tingley & M. Durkin.

C4. Too Large and Too Sparse, or Why the UK Northern Powerhouse Does Not Perform to a London Standard. In *Regional Studies Association Student & Early Career Conference 2016.* Newcastle, UK: 2016, with M. Mayfield & G. Dabinett.

C3. Avoiding Reductionism in Disaster Prevention and the Role of Complexity. In *Vestnick of Almaty University of Power Engineering and Telecommunications Special Issue 2016*, 14–22. Almaty, Kazakhstan: 2016, with G. Punzo & M. Mayfield.

C2. Urban and Rural – An Exploratory Look at Population and Energy Consumption in Local Authorities within England and Wales. In *Regional Urbanism in the Era of Globalisation.* Huddersfield, UK: 2016, with M. Mayfield.

C1. Renewable Energy Technologies in Campus-Sized Developments: A Spatial Study. In *CIBSE Technical Symposium 2015.* London, UK: 2015, with B. R. Lazarov & M. Mayfield.

Funding Awarded

ESRC Impact Acceleration Account — October 2021
£19,225 *– Co-investigator for award allocated by the ESRC-funded IAA to support unlocking productivity in the Sheffield City Region through changes to the mobility network*

Research England Higher Education Innovation Fund — April 2021
£36,942 *– Co-investigator for award allocated by the Research England-funded HEIF to support decarbonization through knowledge exchange partnership with Barnsley Metropolitan Borough Council and South Yorkshire Mayoral Combined Authority*

Research England Higher Education Innovation Fund January 2021
£37,210 *– Co-investigator for award allocated by the Research England-funded HEIF to support the development of a regional emissions accounting and budgeting framework in partnership with the South Yorkshire Mayoral Combined Authority*

EPSRC Impact Acceleration Account January 2021
£9,823 *– Principal investigator and project lead for award allocated by the EPSRC-funded IAA to support the trial of an automated residential retrofit assessment in partnership with Barnsley Metropolitan Borough Council*

Productivity Insights Network Productivity Project Funding March 2020
£9,805 *– Principal investigator and project lead for award allocated by the ESRC-funded Productivity Insights Network to support new and ambitious interdisciplinary directions in productivity research across the social sciences that engage partners and deliver impact* — (Report R1)

Productivity Insights Network Early Career Seed Funding November 2019
£500 *– Award allocated by the ESRC-funded Productivity Insights Network to further develop a research idea presented at the network's early career sandpit event*

Doctoral Supervision

The University of Sheffield

P4. Marcel Fehr 2021 – Present
Energy Retrofit for Commercial Buildings

P3. Danielle Abbey, Primary Supervisor: Danielle Densley Tingley 2021 – Present
Energy Retrofit for Commercial Buildings

P2. Yulan Sheng, Primary Supervisor: Martin Mayfield 2020 – Present
Deep Learning for Urban Thermal Energy Analysis

P1. Will Mihkelson, Primary Supervisor: Danielle Densley Tingley 2019 – Present
Understanding the Relationship Between Resource Consumption and Development Levels

Teaching Activities

Staff Member Responsible, the University of Sheffield

CIV4017/6017 – Computational Fluid Dynamics 2020 – Present
CIV4025/61011 – Thermodynamics for Buildings and Cities 2020 – 2021

Last updated December 22, 2021 — Most artisanally typeset in LaTeX